Two-Chart Slice Obstructions and Binary-Form Rigidity for a Stable-Coordinate Construction

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Abstract

We analyze the explicit affine modification and two-chart locally nilpotent derivation argument in the manuscript *A stable coordinate that is not a coordinate in four variables.* For its auxiliary ring $C = k[a, d, b, c, u]/(ac - bd - 1)$, we classify locally nilpotent derivations fixing $k[a, d, u]$, identify the complete derivative filtration with a Rees algebra, and obtain the exact obstruction ideals $q^N(a, d)^N$ for every order N. Finite Taylor data on the two charts admit an explicit translation law and a terminating global lifting test. A second result extends the source's quadratic rigidity argument to homogeneous binary forms with two distinct projective factors and arbitrary lower auxiliary-degree perturbations; it excludes every homogeneous fiber shift below two. These are local algebraic deductions, not constructions of further stable coordinates. We retain the source's polynomial identities and distinguish them from its filtration, line-bundle lifting, and global noncoordinate argument. Classical slice, factorial-closure, and Rees-algebra tools receive explicit credit; bounded exact checks supplement the proofs. Contribution assessment: C3/5, a source-specific all-order gluing certificate and binary-form rigidity extension, without a new global coordinate theorem.

1 Scope, source architecture, and conventions

All local results below hold over a field k of characteristic zero, unless explicitly stated otherwise. An LND is a k-derivation D for which each element is killed by some power of D; no uniform nilpotence exponent on the whole algebra is assumed. A slice means an element t with $Dt = 1$ in the specified ring, not merely in its fraction field. A coordinate in a polynomial ring is an element belonging to a polynomial generating system.

The source [1] works over $\mathbb{C}$ and asserts that, with

$$Q = x_2^2 - x_4^2 + x_1x_3, \qquad f = x_1 - 2Q\big(Q(x_2 + x_4) + x_1x_4\big),$$

f is not a coordinate of $\mathbb{C}[x_1, x_2, x_3, x_4]$, becomes a coordinate after one polynomial stabilization, and has all fibers isomorphic to $\mathbb{A}^3$. These remain attributed source conclusions. The present report supplies detailed local proofs and a dependency analysis, not a replacement independent certification of the full counterexample.

The cached source contains five scientific sections. We read all their 884 lines, together with the 51-line main file, 129-line bibliography and 18-line README, in bounded literal displays. The resulting 1,082-line reading is a whole-source reading, not a whole-literature or private-corpus proof audit. The source's dependency chain is:

1. `02-construction`, lines 1–200: polynomial presentation, one stabilization, and fiber calculations.

2. `03-filtration`, lines 1–276: an integer-weight filtration, its domain-valued associated graded ring, and extraction of an LND from the hypothesis that p is a coordinate.

3. `04-bundle`, lines 1–233: a punctured line bundle, principalization of an ideal, and lifting an additive action.

4. `05-rigidity`, lines 1–125: a second grading, two independent linear factors, and a two-chart coefficient intersection.

The introduction, lines 1–50, supplies the global claims and context. The parent authenticated the cached subtree in batch44; that is snapshot provenance, not a proof check. Our own before/after hashes are separate.

The fixed-generator ring for our deductions is

$$S = k[a, d, u], \quad K = \mathrm{Frac}(S), \quad C = S[b, c]/(ac - bd - 1), \quad I_0 = (a, d)S. \tag{1}$$

This is a domain: the defining polynomial is primitive and linear in c over the fraction field of $k[a, d, b, u]$, and its coefficients are coprime. The inclusion $S \hookrightarrow C$ is injective, for instance by localizing at a. We use two compatible integer gradings:

	a	d	b	c	u
fiber weight wt	1	1	-1	-1	0
auxiliary degree aux	0	1	-1	0	1

Both make $ac - bd - 1$ homogeneous of degree zero. A derivation of fiber shift e sends C_j into C_{j+e}; "weight-preserving" means $e = 0$. Negative weights are allowed in C, but not in S.

2 The source's explicit construction, kept separate

Here p, s, u, F, J are independent polynomial variables; the capital F is not the binary form used later. Set

$$x = s^2 - u^2 + pF, \qquad H = x^2F - (1 + 2sx)J - pJ^2 - u, \qquad A = k[p, s, u, F, J]/(H). \tag{2}$$

The source's polynomial presentation is based on

$$M = \begin{pmatrix} F & s - u \\ s + u & -p \end{pmatrix}, \quad e = \begin{pmatrix} x \\ -J \end{pmatrix}, \quad n = \begin{pmatrix} J \\ x \end{pmatrix}, \quad M' = (1 - 2ne^t)M.$$

Here $x = -\det M$, $e^t n = 0$, and $H = e^t Me - J - u$. Consequently $(ne^t)^2 = 0$, the displayed matrix transformation has inverse $1 + 2ne^t$, and preserves x and J. If u' is half the difference of its lower-left and upper-right entries, then $u' = u - e^t Me$, so $H = -J - u'$. These identities explain the source's elimination of u' and its presentation $A = k[p', s', F', J]$. In those coordinates

$$x = (s')^2 - J^2 + p'F', \qquad p = p' - 2x(x(s' + J) + p'J).$$

Thus the polynomial in the introduction is indeed the distinguished p under $(x_1, x_2, x_3, x_4) = (p', s', F', J)$. Invertibility here uses preservation of x, J, not just the determinant of a matrix whose entries vary.

For stabilization the source introduces

$$y = s + x(x + u^2), \quad z = sx + pJ, \quad xy - z(z + 1) = p(H + u).$$

In P_p, where $P = k[p, s, u, F, J]$, the variables $p^{\pm 1}, x, y, z, u$ are coordinates. The derivation $\Delta = -p\partial_u$ in these coordinates preserves P, with

$$\Delta p = 0, \quad \Delta u = -p, \quad \Delta s = 2pxu, \quad \Delta J = -2x^2u, \quad \Delta F = -4sxu - 2u, \qquad \Delta H = p. \tag{3}$$

Since the embedding $P \hookrightarrow P_p$ is injective, local nilpotence on P_p implies local nilpotence on P. The complete finite exponential is

$$\begin{aligned} e^{w\Delta}u &= u - pw, \\ e^{w\Delta}s &= s + 2pxuw - p^2xw^2, \\ e^{w\Delta}J &= J - 2x^2uw + px^2w^2, \\ e^{w\Delta}F &= F - (4sxu + 2u)w + p(2sx + 1 - 4x^2u^2)w^2 \\ &\qquad + 4p^2x^2uw^3 - p^3x^2w^4. \end{aligned} \tag{4}$$

This is an explicit polynomial automorphism with inverse obtained by $w \mapsto -w$; it carries H to $H + pw$ and fixes p.

For completeness, put $x_0 = s^2 - u^2$ and

$$L = x_0^2F - (1 + 2sx_0)J, \qquad N = (1 - 2sx_0)F + 4s^2J.$$

The determinant of this F, J change is one, with inverse

$$F = 4s^2L + (1 + 2sx_0)N, \qquad J = -(1 - 2sx_0)L + x_0^2N.$$

Moreover

$$H = L - u + pQ_0, \qquad Q_0 = 2x_0F^2 + pF^3 - 2sFJ - J^2.$$

Changing w to $w + Q_0$ in $P[w]/(H + pw)$ leaves the equation $L - u + pw = 0$, which eliminates L. This is the source's explicit stabilization, not a consequence of our later slice-obstruction theorem. The zero fiber follows from $H \bmod p = L - u$; the other fibers follow by solving $u = (xy - z(z + 1))/p$ at a fixed nonzero value of p.

3 The exact two-chart intersection and derivations

The local shear derivation is

$$D(a) = D(d) = D(u) = 0, \qquad D(b) = a, \qquad D(c) = d. \tag{5}$$

It is locally nilpotent because it lowers polynomial degree in b, c. Its fiber shift is two and its auxiliary shift is one.

Lemma 3.1 (Intersection with the invariant field)**.** *Inside* $\mathrm{Frac}(C)$ *one has* $C \cap K = S$.

Proof. The two presentations are

$$C_a = S_a[b], \quad c = (1 + bd)/a; \qquad C_d = S_d[c], \quad b = (ac - 1)/d.$$

If an element of either polynomial ring is in K, its polynomial variable coefficients of positive degree vanish. Hence $C \cap K \subseteq S_a \cap S_d$. A reduced denominator in the UFD S must then divide both a power of a and a power of d; it is a unit. The reverse inclusion is immediate. □

Theorem 3.2 (Classification with three fixed generators)**.** *Every S-derivation of C is uniquely qD for some $q \in C$. It is locally nilpotent if and only if $q \in S$. For $q \neq 0$ its kernel is exactly S. If qD is homogeneous of fiber shift e, then q is homogeneous of weight $e - 2$; in particular every nonzero such LND has $e \geq 2$.*

Proof. For an S-derivation δ, differentiating the relation gives $a\delta(c) = d\delta(b)$. Define $q = c\delta(b) - b\delta(c)$. Then $aq = (ac - bd)\delta(b) = \delta(b)$ and $dq = \delta(c)$, proving the assertion and uniqueness in the domain C.

If δ is an LND, it extends to an LND of $C \otimes_S K = K[b]$, since all denominators are in its kernel. A derivation $P(b)\partial_b$ of $K[b]$ is locally nilpotent only when P is constant: if $\deg P = r \geq 1$, repeated application to b has nonzero leading coefficient and degree $1 + j(r-1)$ after j applications. Characteristic zero prevents the factors of the leading coefficient from vanishing. Thus $aq = \delta(b) \in K$, and Lemma 3.1 gives $q \in S$. Conversely, $q \in S = \ker D$ implies $(qD)^j = q^j D^j$, so qD is locally nilpotent. A nonzero such derivation has kernel K after extension to $K[b]$, hence kernel S in C. Finally $\delta(b) = aq$ has weight $e-1$, forcing $\mathrm{wt}(q) = e-2$. □

This theorem classifies derivations *already fixing* a, d, u. It does not say that every LND on C fixes them. For example ∂_u is an LND of fiber shift zero. Nor does it compute the Makar–Limanov invariant of C by classifying only this subfamily.

4 All-order obstruction ideals and a Rees algebra

Let $\mathcal{F}_{-1} = 0$ and $\mathcal{F}_N = \ker D^{N+1}$ for $N \geq 0$. These are S-modules, not finite-dimensional k-spaces. The following strengthens a chart intersection into a complete degree-by-degree certificate.

Theorem 4.1 (Exact derivative filtration)**.** *For every integer $N \geq 0$, $\mathcal{F}_N$ consists exactly of classes of polynomials of total b, c degree at most N over S, and*

$$0 \longrightarrow \mathcal{F}_{N-1} \longrightarrow \mathcal{F}_N \xrightarrow{D^N/N!} I_0^N \longrightarrow 0 \tag{6}$$

is exact, with $I_0^0 = S$. The induced graded algebra identification is

$$\mathrm{gr}_{\mathcal{F}} C \simeq \bigoplus_{N \geq 0} I_0^N T^N = S[aT, dT] \subseteq S[T]. \tag{7}$$

For $0 \neq q \in S$, the filtration for qD is the same and $(qD)^N(\mathcal{F}_N)/N! = q^N I_0^N$.

Proof. For a homogeneous polynomial $G(B, C)$ of degree N over S, substituting $b + at, c + dt$ shows that $D^N G(b, c)/N! = G(a, d)$. The homogeneous kernel of

$$S[B, C] \longrightarrow S[T], \qquad B \mapsto aT,\ C \mapsto dT$$

is $(aC - dB)$. Indeed over K the kernel is the same linear equation; $aC - dB$ is primitive over the UFD S, so Gauss's lemma contracts its principal ideal back to $S[B, C]$.

Take a representative of minimal total b, c degree r for an element $f \in C$. If $r > 0$ and its top part G_r satisfies $G_r(a, d) = 0$, write $G_r = (ac - db)Q_{r-1}$. Subtract $(ac - db - 1)Q_{r-1}$ from the representative. This preserves f and lowers its degree, a contradiction. Therefore $D^r f/r! = G_r(a, d) \neq 0$, while $D^{r+1} f = 0$. For $r = 0$ the same conclusion is immediate for nonzero f. This proves equality of minimal representative degree and D-degree.

The kernel in (6) is now $\mathcal{F}_{N-1}$. Its image is I_0^N, since $D^N(b^i c^{N-i})/N! = a^i d^{N-i}$ for $0 \leq i \leq N$. The divided top derivative of a product is the product of its divided top derivatives, by the Leibniz formula. Thus the module identifications assemble into (7). Finally $(qD)^j = q^j D^j$ and C is a domain, proving the statement for replicas qD. □

Corollary 4.2 (Plinth ideal and absence of a global slice)**.** *For $0 \neq q \in S$,*

$$\mathrm{pl}(qD) := (qD)(C) \cap \ker(qD) = q(a, d)S. \tag{8}$$

In particular no qD has a global slice, although $(a, d)C = C$.

Proof. If $(qD)f$ is invariant, then $(qD)^2 f = 0$, so Theorem 4.1 with $N = 1$ applies. The ideal $q(a, d)$ is contained in the proper ideal (a, d) of S, and cannot contain one. In C, however, $ac - bd = 1$. □

The term plinth ideal and the associated local bundle construction are classical; see [2], Definition 5.1. The Rees algebra of a two-generated regular-sequence ideal is also classical. The useful increment here is the exact all-order ideal in the source's own variables, including replicas, rather than interpreting a finite matrix rank as a global slice certificate. No new general invariant-theory theorem is claimed.

As an explicit second-order example, $g = b^2 + ubc + c^2$ satisfies

$$D^2 g/2 = a^2 + uad + d^2, \qquad D^3 g = 0.$$

Every prescribed $h \in (a, d)^2$ is similarly the divided second derivative of a polynomial of degree at most two. A prescribed unit is not such a derivative in $\mathcal{F}_2$. More generally membership in I_0^N is tested by the absence of monomials $a^i d^j u^r$ with $i + j < N$.

5 Finite Taylor data, transition, and exact lifting

On the two charts define

$$t_a = b/a, \qquad t_d = c/d, \qquad h = 1/(ad).$$

Then $Dt_a = Dt_d = 1$, $t_d = t_a + h$, and $C_a = S_a[t_a]$, $C_d = S_d[t_d]$. Evaluation at the zero slice is the finite operator, on any element on which D is nilpotent,

$$\Pi_a f = \sum_{j \geq 0} \frac{(-t_a)^j}{j!} D^j f. \tag{9}$$

It sends b to zero and c to $1/a$, and fixes S_a. Thus it is a ring homomorphism onto S_a. The corresponding Π_d sends c to zero and b to $-1/d$. These maps are not defined on the missing denominator locus by formal substitution. They are not orthogonal projections for an unspecified Hilbert-space norm.

Proposition 5.1 (Exact chart data)**.** *If $f \in \mathcal{F}_N$, then*

$$f = \sum_{j=0}^{N} A_j t_a^j = \sum_{j=0}^{N} B_j t_d^j, \quad A_j = \frac{\Pi_a D^j f}{j!}, \quad B_j = \frac{\Pi_d D^j f}{j!},$$

where $A_j \in S_a$, $B_j \in S_d$. The transition is exactly

$$A_j = \sum_{\ell=j}^{N} \binom{\ell}{j} B_\ell h^{\ell-j}, \qquad B_j = \sum_{\ell=j}^{N} \binom{\ell}{j} A_\ell (-h)^{\ell-j}. \tag{10}$$

Conversely any two such polynomial data satisfying (10) come from a unique $f \in \mathcal{F}_N$. In particular $A_N = B_N \in I_0^N$.

Proof. On either chart D is differentiation in its slice coordinate. Ordinary finite polynomial Taylor expansion proves the analysis and synthesis formulas, and substitution $t_d = t_a + h$ gives the binomial transition and its inverse. Compatible data define an element of $C_a \cap C_d$. This intersection equals C inside its fraction field: if $a^r f, d^s f \in C$ with $r, s \geq 1$, expand $(ac - bd)^{r+s-1} = 1$. Every term is divisible by either a^r or d^s in C, so $f \in C$. The resulting element is killed by D^{N+1} in both charts and hence in C. Its top coefficient is $D^N f/N!$, so Theorem 4.1 gives the last assertion. □

There is also a one-chart algorithm that does not presuppose a compatible second chart. Given $f_a \in S_a[t_a]$ of degree at most N, repeat the following operation on its current top degree r. Test whether its leading coefficient H_r belongs to $I_0^r \subset S$. If it does, write

$$H_r = \sum_{i=0}^{r} s_i a^i d^{r-i}, \qquad s_i \in S,$$

subtract $\sum_i s_i b^i c^{r-i}$ in the a-chart, and continue at smaller degree. At degree zero require membership in S. This terminates after at most $N+1$ coefficient tests. It succeeds if and only if $f_a \in C$. Necessity at each stage follows from (6); sufficiency follows by adding the exhibited global lifts. Different choices of s_i differ by lower-degree global elements, so they do not affect success. Membership and decomposition are exact polynomial operations, not numerical rank decisions. Merely testing the top coefficient is insufficient: $b^N + 1/a$ passes the top test for $N \geq 1$ and fails the constant test.

For $g = b^2 + ubc + c^2$, the a-chart expansion is

$$g = (a^2 + uad + d^2)t_a^2 + (u + 2d/a)t_a + a^{-2}.$$

The nonpolynomial-looking lower coefficients are allowed; the polynomial g supplies their exact compatibility. By contrast t_a has top coefficient one, so cannot be a global element of $\mathcal{F}_1$.

A direct first-order obstruction makes the distinction especially clear. If $Df = H \in S$, then on the charts $f = Ht_a + r_a = Ht_d + r_d$, with $r_a \in S_a, r_d \in S_d$. Hence

$$H/(ad) = r_a - r_d \quad \text{in } S_{ad}. \tag{11}$$

In the Laurent expansion, $S_a + S_d$ has no monomial with both a, d exponents negative. The coefficient of $a^{-1}d^{-1}$ on the left is $H(0,0,u)$. Therefore (11) is soluble exactly when $H \in (a,d)$. For $H = aA + dB$, the explicit solution $f = Ab + Bc$ proves sufficiency. This is the elementary Cech explanation of (8). The quotient map has image $\operatorname{Spec} S \setminus V(a,d)$, as the two chart presentations show. Its missing locus in $\operatorname{Spec} S$ must not be confused with a missing locus in $\operatorname{Spec} C$: the latter is covered by $D(a)$ and $D(d)$.

6 Binary-form extension of the rigidity step

We first record the two standard facts needed for the argument, so that the quantifiers in the extension do not rest on a finite computation.

Lemma 6.1 (Factorial closure and highest homogeneous part)**.** *The kernel of an LND on a characteristic-zero domain is factorially closed: if $rs \neq 0$ and $D(rs) = 0$, then $Dr = Ds = 0$. For a finitely generated integer-graded domain, the highest nonzero homogeneous part of an LND is a nonzero homogeneous LND.*

Proof. Apply the automorphism $\exp(tD)$ to rs. The degrees in t of the two nonzero polynomials $\exp(tD)r$ and $\exp(tD)s$ add. Their product is constant, so both have degree zero. For the second assertion, finitely many homogeneous algebra generators bound the shifts of D above and below. Taking homogeneous components of the Leibniz identity gives derivations, with a highest nonzero one D_r. For a homogeneous element, D_r^j is the component of D^j in shift jr. Since D^j eventually vanishes on that element, so does D_r^j. There are only finitely many components of any element, completing the proof even when $r < 0$. □

Theorem 6.2 (Two projective factors and the shift barrier)**.** *Let $m \geq 2$ and let $F_m(X,Y) \in k[X,Y]$ be a nonzero homogeneous form of degree m with at least two distinct projective linear*

factors over $\overline{k}$. Let $R \in C$ be homogeneous of fiber weight m, all of whose auxiliary homogeneous components have degree strictly less than m. Set

$$v = F_m(au, d) + R. \tag{12}$$

If a fiber-homogeneous LND δ of shift $e < 2$ satisfies $\delta v = 0$, then $\delta = 0$. In particular no nonzero weight-preserving LND kills v. The assertion includes $R = a^{m+1}b$ and allows repeated factors of F_m.

Proof. Assume $\delta \neq 0$. First extend scalars to $\overline{k}$. The same ring presentation is a domain there, and a nonzero derivation stays nonzero; thus it suffices to derive a contradiction after this extension. The elements au, d are algebraically independent, as follows from the inclusion $k[a, d, u] \subset C$. The top auxiliary part of v is consequently the nonzero polynomial $F_m(au, d)$, of degree m.

Let δ' be the highest auxiliary part of δ. By Lemma 6.1, it is a nonzero LND; since the two gradings are compatible it retains fiber shift e. The highest component of $\delta v = 0$ gives $\delta' F_m(au, d) = 0$. Factor the form into linear factors over $\overline{k}$. Repeated factorial closure shows that every one is fixed by δ'. Two distinct projective factors are two independent linear forms in au, d, so $\delta'(au) = \delta' d = 0$. Factorial closure applied to au gives $\delta' a = \delta' u = 0$. Theorem 3.2, now over $\overline{k}$, says $\delta' = qD$ with $q \in \overline{k}[a, d, u]$ of fiber weight $e - 2 < 0$. There is no nonzero element of that weight in this polynomial ring. This contradicts $\delta' \neq 0$. □

The source is the case $m = 2$, $F_2 = X^2 - Y^2$, $R = a^3 b$, $e = 0$. The theorem permits, for example,

$$v = (au - d)^2(au + 2d) + a^4 b$$

at $m = 3$. Repeated factors do not obstruct the proof; a single projective direction does. Indeed $F_m = Y^m$, $R = a^{m+1}b$ is killed by the nonzero weight-preserving LND ∂_u. The shift threshold is also sharp for the stated family: for $R = 0$, the nonzero derivation D of shift two kills every $F_m(au, d)$. These examples validate the need for the hypotheses; they do not classify all LNDs at shifts $e \geq 2$ for general R.

This extension concerns the auxiliary ring and the distinguished element only. For general m, F_m, R we have proved neither a polynomial presentation analogous to A, nor a stable automorphism, nor a principalization of the source's base ideal. In particular it is not a family of new stable noncoordinates. The parent suggested the binary-factor extension; the proof and fixed-generator/all-order analysis here are author-checked.

7 What the global source argument additionally requires

We spell out the connection to the source rather than replacing its geometry with a matrix computation. In `03-filtration` the weights of (p, s, u, F, J) are $(-1, 0, 0, 1, 1)$. The source proves separatedness and identifies the associated graded domain as

$$G = k[p, s, u, F, J]/(H^*), \qquad H^* = H + u.$$

Writing τ for the initial form of p, it describes

$$B = k[x, y, z, u]/(xy - z(z + 1)), \qquad G_\tau = B[\tau, \tau^{-1}],$$

with B in degree zero. With

$$s_B = y - x(x + u^2), \quad f_0 = x - s_B^2 + u^2, \quad g_0 = z - s_B x, \quad I = (f_0, g_0) \subset B,$$

the source obtains $G = B[\tau, I/\tau]$ and $G_\ell = \tau^{-\ell} I^\ell$ for $\ell > 0$. This is a different filtration and a different ideal from $\mathcal{F}$ and I_0 in Section 4; conflating them would bypass an essential part of the source proof.

Under the hypothesis that p is a coordinate, the source chooses polynomial coordinates on A and a partial derivative killing p and a suitable positive-degree coordinate. Its initial derivation is a nonzero LND on G. After localizing at τ it has the form $\tau^{-r}E$ on B, where E is a nonzero LND of B with a nonzero invariant in some $I^{\ell} \subset I$. The choice of a positive-degree fixed element, nonvanishing of the initial derivation, and its descent to B are genuine extraction requirements. They do not follow from $C \cap K = S$.

The next source section embeds B into C by

$$x = ab, \qquad y = dc, \qquad z = db, \qquad u = u.$$

Here $xy = z(z+1)$ follows from $ac - bd = 1$. For $v = a^3b + a^2u^2 - d^2$, the source proves $IC = (v)$. On the a-chart it sets $\xi = d/a$, $j = x + u^2 - \xi^2$ and computes

$$s_B = \xi - xj, \quad f_0 = (1 + 2\xi x - x^2 j)j, \quad g_0 = x^2 j, \quad v = a^2 j.$$

The explicit Bezout identity is

$$(1 - 2\xi x)(1 + 2\xi x - x^2 j) + x^2(4\xi^2 + j - 2\xi x j) = 1.$$

At primes containing a, one has $bd = -1$, $g_0 = -1$ and $v = -d^2$, so the missing chart cannot be discarded. The source then uses its punctured line-bundle description, invariance of its Picard group under adjoining an affine-line parameter, and normalization of the linearization to lift E to a nonzero fiber-weight-preserving LND of C. An invariant nonzero element of I becomes divisible by v; factorial closure forces the lift to kill v.

Theorem 6.2 now supplies exactly the terminal contradiction for the source's v. This local endpoint has been checked and extended. The finite tests below check polynomial identities and bounded algebra; they do not verify the Picard argument, the extraction for every possible coordinate system, or an arbitrary-LND classification. No source algebraic error was identified in the checked identities. The needed precision is in their scope: a chart slice is not a global slice, the two filtrations are distinct, and stabilization is a separate construction.

8 Private proof transfer, literature, and limits

The fresh private reading had two deliberately different roles. TR-item-061-astra, lines 290–370, contains the finite exponential local slice proof and a denominator-saturation criterion; lines 1–80 locate its scope. It is a later comparative report, not by itself the original Ram derivation. The original TR-rev-1-old-tr250 was read at lines 140–180 and 435–476. Its finite polynomial moment argument proves injectivity on a declared finite physical space, and its Gram proof uses positivity on that space. We use the discipline of specifying the space and the inverse on its range, not its differential-equation conclusions.

In the present algebra the corresponding analysis is the finite Taylor map $f \mapsto (A_0, \ldots, A_N)$ on $\mathcal{F}_N$, and synthesis is $\sum A_j t_a^j$. This is over the localized coefficient ring S_a, not an invertible Gram matrix on all of C. Global realizability is the extra condition supplied by (10) or the recursive ideal test. This produces an actual denominator/gluing constraint rather than merely renaming a vector of coefficients. The private local-slice proof also explains why denominator removal requires a separate saturation argument. Only the stated 244 private lines were freshly read; unread complements are retained in the audit, and the private corpus stayed local.

Public-primary searches were refreshed on October 8, 2026, using public titles and public mathematical terms. Blanc–Poloni [3] study affine-plane bundles over the punctured affine plane and bivariables, including the second Vénéreau polynomial. Dutta–Lahiri [4] study residual and stable coordinates under hypotheses on the coefficient ring. These are relevant neighboring results, not independent verification of the source's specific degree-five polynomial. The contemporary survey [5] has a February 17, 2026 revision; its abstract concerns the reach and limitations

of LND methods for cancellation. We checked that metadata/abstract and the classical slice statement in its first-version Section 1.3, not the entire survey or all its claimed generalizations. The familiar plinth framework is documented in the primary article [2], Section 5.2. None of these scoped reads is a claim that the global stable-coordinate question has been independently settled by the literature review.

Advantages. The classification has no numerical conditioning assumptions; the all-order obstruction is a monomial ideal with a direct membership test. The chart transition has exact signs and factorial normalization. The binary theorem accommodates repeated factors, nonsplit forms by scalar extension, and every perturbation below the top auxiliary degree.

Limitations. The filtration modules are not finite-dimensional over k, and the algorithm is finite only for a specified input degree. Characteristic zero is essential for the derivative and factorial-closure arguments. No analytic norm or uniformly bounded coefficient height is asserted. The known $SL_2/\mathbb{G}_a$ torsor structure, local slices, plinth ideal, and Rees machinery are not discoveries of this report. The specific extension does not establish stable-coordinate examples for higher binary degree, nor does it replace the source's geometric lifting.

Feasible next steps. One can implement the recursive ideal test with sparse exact polynomials, recording actual denominator and coefficient growth. Another precise problem is to determine which binary-form perturbations admit the source's principalization and polynomial-presentation steps. Those are additional algebraic problems, not consequences promised by the local rigidity theorem. A broader classification of LNDs on C must allow them to move a, d, u and cannot start by assuming the conclusion of Theorem 3.2.

9 Verification and assessment

The owned script `_audit/M109-checks.py` performs exact rational polynomial checks of the construction, divided derivatives, slice reconstruction, binomial transitions, leading-form relations, and the degree hypotheses and boundary examples in Section 6. Its JSON records named groups, bounds, and actual assertion counts. The author run passed 3,129 exact assertions, including 1,620 divided derivative identities and 455 binomial transition inverse entries. No floating-point tolerance is used. These tests support normalization and algebraic bookkeeping; the all-degree statements rest on the proofs. Parent tests are separate evidence and are not included in the author's counts or frozen as owned files.

The read ledger records literal line intervals and their exact unread complements, with hashes captured before and after reading. The proof ledger separates source formulas, proved local extensions, classical tools, and global dependencies. Native compiler diagnostics, the export log, and all-page visual review are retained separately. A final manifest freezes only this report and its owned evidence/build outputs.

The contribution is assessed as C3, independently of catalogue applicability L3. The basis is the exact all-order obstruction and terminating global lifting certificate, together with the binary-form extension and sharp homogeneous-shift boundary. Elementary identities, the classical torsor interpretation, and the bounded tests are not counted as independent discoveries. No global theorem certification, claim of first-in-literature priority, or new cancellation resolution is included in that assessment.

References

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[2] E. Dufresne and H. Kraft, *Invariants and separating morphisms for algebraic group actions*, Math. Z. 280 (2015), 231–255, Section 5.2, Definition 5.1 and Proposition 5.2. Author-hosted primary article.

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[6] Private local corpus, TR-item-061-astra, literal lines 1–80 and 290–370, read locally; local-slice and saturation proof comparison. Not a public literature source.

[7] Private original Ram corpus, TR-rev-1-old-tr250, literal lines 140–180 and 435–476, read locally; finite-space moment inversion and polynomial Gram proof. No online transfer.